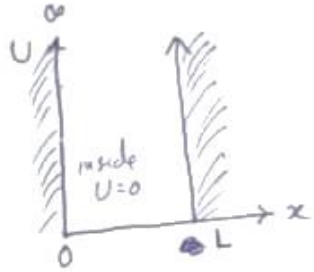
I. Particle in a Box : Beiser

(a) Particle trapped in a box with infinitely hard walls



- Particle is restricted along x-axis : between $x=0$ and $x=L$.
- Particle does not lose energy when it collides with such walls, so that its total energy stays const.
- Inside U is constant, can be zero. $U=0$ inside
  outside at both ends, U is infinite.

$$\psi \rightarrow \psi \quad \text{for } x \geq 0,\ x > L$$

$$\rightarrow 0 \quad \text{for } x \geq L \text{ and } x \leq 0$$

with the box

$$\frac{d^2\psi}{dx^2} + \frac{2mE}{\hbar^2}\psi = 0 \quad \rightarrow (1)$$

$$\frac{p^2}{2m} + U = E$$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + U\psi = E\psi$$

stationary state

$U=0 \rightarrow$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi$$

or $$\frac{d^2\psi}{dx^2} = -\frac{2mE}{\hbar^2}E\psi$$

$$\frac{d^2\psi}{dx^2} + \frac{2mE}{\hbar^2}E\psi$$

Solution for (1)

$$\psi = A\sin\left(\frac{\sqrt{2mE}}{\hbar}\right)x + B\cos\left(\frac{\sqrt{2mE}}{\hbar}\right)x$$

But $\psi = 0$ at $x=0$ $\rightarrow B=0$ as $\cos x = 1$ at $x=0$

$$\psi = A\sin\frac{\sqrt{2mE}}{\hbar}x$$

But $\psi = 0$ at $x=L$ means $\frac{\sqrt{2mE}}{\hbar}L = n\pi$

$$\boxed{E_n = \frac{n^2\hbar^2}{8mL^2}}$$

$$\frac{\sqrt{2mE}}{\hbar}x = n\pi, \quad n = 1, 2, 3, \ldots$$

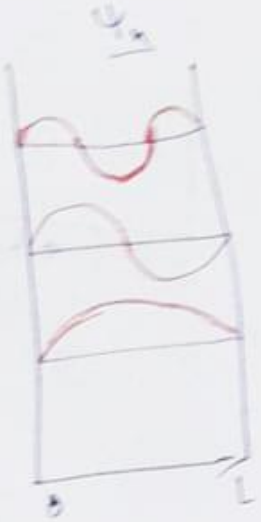
at $x=L$

Energy levels, $$\boxed{E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}} = \frac{n^2\pi^2\hbar^2}{4\pi^2 2mL^2} = \frac{n^2h^2}{8mL^2}$$

$$\boxed{\frac{\sqrt{2mE_n}}{\hbar}L = n\pi}$$

$$\boxed{E_n = \frac{n^2\pi^2\hbar^2}{2mL^2} = \frac{n^2h^2}{8mL^2}} \quad \boxed{\psi_n = A\sin\frac{n\pi x}{L}}$$

$$\psi_n = \sqrt{\frac{2}{L}}\sin\frac{n\pi x}{L}, \quad n = 1, 2, 3$$



$$\int_{-\infty}^{\infty}\psi^*\psi\,dx = 1$$

$$\int_0^L \psi^*\psi\,dx = 1$$

$$A^2\int_0^L \sin^2\left(\frac{n\pi x}{L}\right)dx =$$

$$\frac{A^2}{2}\left[\int_0^L dx - \int_0^L \cos\left(\frac{2n\pi x}{L}\right)dx\right]$$

$$\frac{A^2}{2}\left[x - \left(\frac{L}{2n\pi}\right)\sin\frac{2n\pi x}{L}\right]$$

$$= A^2\left(\frac{L}{2}\right) = 1$$

or $A = \sqrt{2/L}$

$|\psi_1|^2$ has max $2/L$ at middle.

$n=1$ → most likely at $L/2$

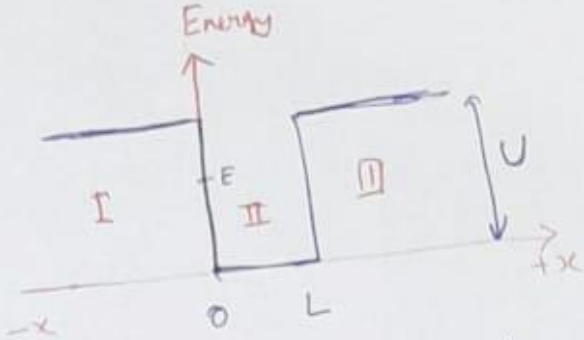
$n=2$ → never at $L/2$

Finite Potential Well

Particle in a well → Energy < U

CM → Particle does not enter in I, III

QM → Particle has a certain probability that it can penetrate in regions I and III



A square potential well with finite barriers. $E < U$.

Region I & Region III

$E < U$, Energy of Particle → $(U-E)$

$$\frac{p^2}{2m} + U = E \Rightarrow -\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + U\psi = E\psi$$

$$-\frac{d^2\psi}{dx^2} = \frac{2m}{\hbar^2}(E-U)\psi$$

$$\frac{d^2\psi}{dx^2} = -\frac{2m}{\hbar^2}(E-U)\psi$$

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E-U)\psi = 0$$

or $$\frac{d^2\psi}{dx} - \frac{2m}{\hbar^2}(U-E)\psi = 0 \qquad \left(\text{as } U > E \text{ or } E < U\right)$$

Let $a = \frac{\sqrt{2m(U-E)}}{\hbar}$ or $a^2 = \frac{2m(U-E)}{\hbar^2}$

Region I and III $\boxed{\frac{d^2\psi}{dx^2} - a^2\psi = 0}$ for $x < 0$ & $x > L$

→ gives real exponents

$$\psi_I = Ae^{ax} + Be^{-ax}$$

$$\psi_{III} = Ce^{ax} + De^{-ax}$$

I	II	III
U	E<U	U

Regions I & III : PE = U but U > E

$$\frac{d^2\psi}{dx^2} - a^2\psi = 0 \quad , \quad a^2 = \frac{\sqrt{2m(U-E)}}{\hbar}$$

$$\psi_I = Ae^{ax} + Be^{-ax} \quad \text{and} \quad \psi_{III} = Ce^{ax} + De^{-ax}$$

Region II

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}E\psi = 0 \quad \text{or} \quad U = 0 \text{ here}$$

$$\psi_{II} = F\sin\frac{\sqrt{2mE}}{\hbar}x + G\cos\frac{\sqrt{2mE}}{\hbar}x$$

(a) Region I and Region II

- All regions → $\psi_I, \psi_{II}, \psi_{III}$ need to be continuous and finite

But $e^{-ax} \to \infty$ as $x \to -\infty$ ⇒ coefficient $B = 0$

$e^{ax} \to \infty$ as $x \to \infty$ ⇒ coefficient $C = 0$

∴ Region I → $\psi_I = Ae^{ax}$ Region III → $\psi_{III} = De^{-ax}$

ψ_I and ψ_{III} decrease exponentially inside the barriers at the sides of the well.

Within the well

$$\psi_{II} = F\sin\frac{\sqrt{2mE}}{\hbar}x + G\cos\frac{\sqrt{2mE}}{\hbar}x$$

Compare with → infinite potential well, $\psi = 0$ at $x = 0$ & $L = 0$.

But here, it is not!

Here, $\psi_{II} = A$ at $x = 0$ and $\psi_{II} = D$ at $x = L$

Both sine and cosine exists.

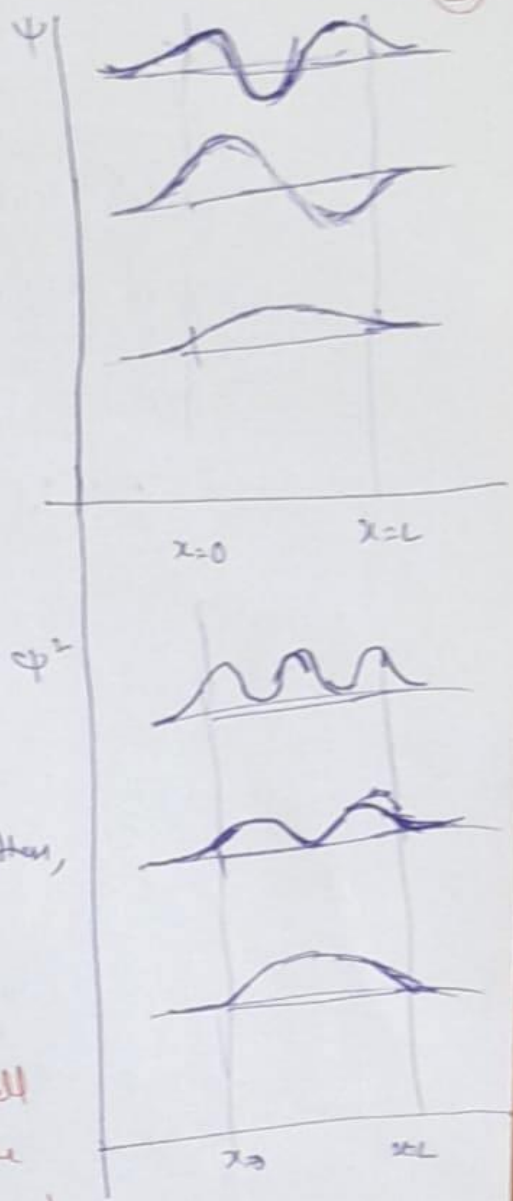
Finite Potential well

$$\psi_I = A e^{ax}$$

$$\psi_{II} = F \sin \frac{\sqrt{2mE}}{\hbar} x + G \cos \frac{\sqrt{2mE}}{\hbar} x$$

$$\psi_{III} = D e^{-ax}$$

ψ, $\frac{d\psi}{dx}$ must be continuous at $x=0$ and $x=L$: the wave functions inside and outside each side of the well must not only have same value ~~but also~~ where they join, but also the same slopes, so that they match up perfectly. Because of this, exact matching occurs for certain specific values E_n of the particle energy.

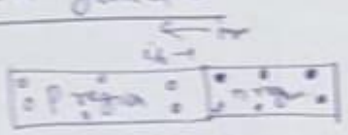


Because the wavelengths that fit into well are longer than for an infinite well of the same width, the corresponding particle momenta are lower ($\lambda = h/p$). Hence the energy levels E_n are lower for each n than they are for a particle in an infinite well

Tunnel Effect :

"A particle without the energy to pass over a potential barrier may still tunnel through it".

P-n junction

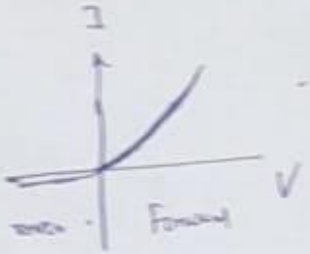


P-n junct.

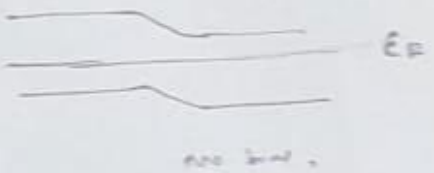
- bce $\bar{e}$
- o vle

i_d → Normal $\bar{e}$ current

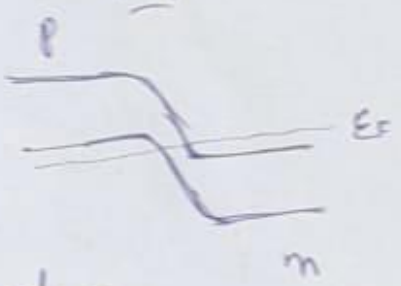
i_r → recombination $\bar{e}$ current.



PN semiconductor



n-p

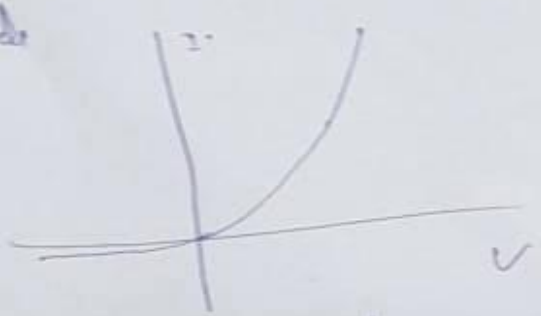


In tunnel diode : depletion region is small

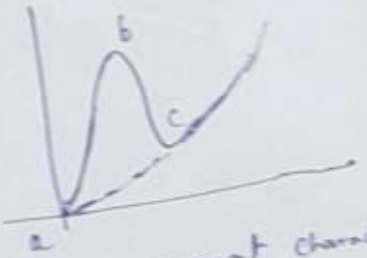
Because the depletion region is so narrow, only a few electron wandering across, electrons can tunnel through the forbidden band. ~~these~~

When no external voltage is applied to the diode, $\bar{e}$ tunnels in both directions across the gap in equal numbers and the Fermi energy is constant across the diode.

At higher ~~voltage~~ voltage. Tunnel diode acts as normal diode



regular diode



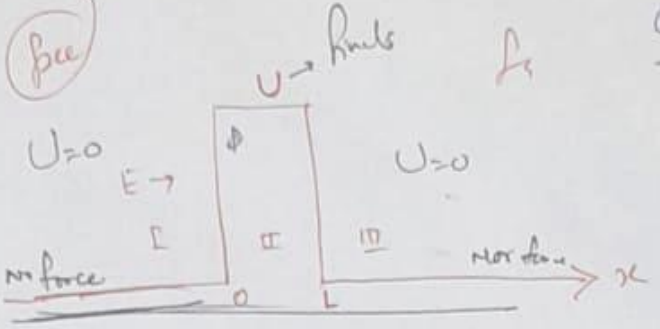
V-I current characteristic of a tunnel diode. the pts a, b, c

Tunnel diode → voltage change rapidly between a or b or b e c

Respond 11 quicker → are used in high freq oscillators → as fast switches in computer

Tunnel effect

(Bce)



(I) $\frac{d^2\psi_I}{dx^2} + \frac{2m}{\hbar^2} E\psi_I = 0$

$\frac{d^2\psi_{III}}{dx^2} + \frac{2m}{\hbar^2} E\psi_{III} = 0$

In CM: when a particle of energy $E < U$, approaches a potential barrier, according to cm, the particle must be reflected.

In QM: In QM, the de Broglie waves that correspond to the particle are partly reflected and partly transmitted, which means that the particle must has a finite chance of penetrating barrier

$$\psi_I = Ae^{ik_1x} + Be^{-ik_1x}$$

$$\psi_{III} = Fe^{ik_1x} + Ge^{-ik_1x}$$

$$k_1 = \frac{\sqrt{2mE}}{\hbar} = \frac{p}{\hbar} = \frac{2\pi}{\lambda}$$ ← out side barr.

Incoming wave: $\psi_{I+} = Ae^{ik_1x}$

Reflected wave: $\psi_{I-} = Be^{-ik_1x}$

$$\psi_I = \psi_{I+} + \psi_{I-}$$

Transmitted

$\psi_{II} = \psi_{III+} = Fe^{ik_1x}$ $\quad \psi_{III-} = 0$ as → $G=0$.

Transmission probability: $T \propto \frac{|\psi_{III+}|^2}{|\psi_{I+}|^2} \propto \frac{FF^*}{AA^*}$

"T" is the fraction of incident particles that succeed tunnelling through the barrier.

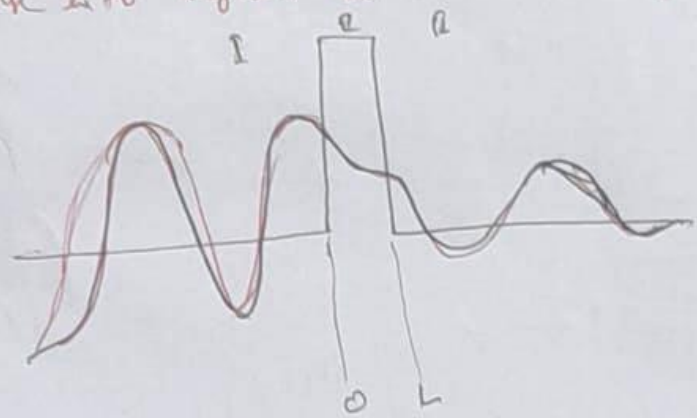
Region II : Since $E < U$

$$\frac{d^2\psi_{II}}{dx^2} - \frac{2m}{\hbar^2}(U-E)\psi_{II} = 0$$

$k_2 = \frac{\sqrt{2m(U-E)}}{\hbar}$ wave number inside barrier.

$$\boxed{\psi_{II} = Ce^{-k_2 x} + De^{k_2 x}}$$

Since exponents are real quantities, ψ_{II} does not oscillate and so does not represent a moving particle. But $|\psi_{II}| \neq 0$, so there is a finite probability of finding a particle within a barrier! Such particle may emerge into region III or it may return to region I.



Applying Boundary conditions

Transmission Probability, $T = \left[\frac{16}{4 + (k_2/k_1)^2}\right] e^{-2k_2 L}$

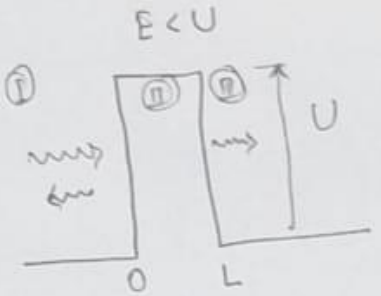
$$\left(\frac{k_2}{k_1}\right)^2 = \frac{2m(U-E)/\hbar^2}{2mE/\hbar^2} = \frac{U-E}{E} = \frac{U}{E} - 1$$

i.e. $T = XY = \underbrace{\left[\frac{16}{4+(k_2/k_1)^2}\right]}_{X} \underbrace{e^{-2k_2 L}}_{Y}$

X varies less with E and U than does the exponential

$X \approx 1$ in value. $\boxed{T \approx e^{-2k_2 L}}$ ✓

Tunnelling



$$\left.\begin{aligned} \frac{d^2\psi_I}{dx^2} + k_1^2\psi = 0 \\ \frac{d^2\psi_{III}}{dx^2} + k_1^2\psi_{III} = 0 \end{aligned}\right\} \quad \boxed{k_1 = \frac{\sqrt{2mE}}{\hbar}} \text{ and } k_2 = \frac{\sqrt{2m(U-E)}}{\hbar}$$

$$\frac{d^2\psi_{II}}{dx^2} - \frac{2m(U-E)}{\hbar^2}\psi_{II} = 0 \quad \text{or} \quad \frac{d^2\psi_{II}}{dx^2} - k_2^2\psi_{II} = 0$$

Transmission Probability $= T = \left[\frac{16}{4+(k_2/k_1)^2}\right] e^{-k_2 L}$

≈ 1

$$\boxed{T \approx e^{-k_2 L}}$$

$$\boxed{T \approx e^{-\frac{\sqrt{2m(U-E)}}{\hbar} L}}$$

Problem:

Electrons with energies 1.0 eV and 2.0 eV are incident on a barrier 10.0 eV high and 0.5 nm wide.

a) Find their respective transmission probabilities

b) How are these affected if the barrier is doubled in width?

a) For 1.0 eV electron,

$$k_2 = \frac{\sqrt{2m(U-E)}}{\hbar} = \frac{\sqrt{2(9.1\times10^{-31}\,kg)[(10-1)eV(1.6\times10^{-19}\,J/eV)]}}{1.054\times10^{-34}\,J.s}$$

$$k_2 = 1.6\times10^{10}\ /m$$

Since $L = 0.5$ nm, $2k_2L = 2\times1.6\times10^{10}\,m^{-1}\times5\times10^{-10}\,m = 16$

$$\therefore T = e^{-2k_2L} = e^{-16} = 1.1\times10^{-7}$$

So, one 1.0 eV $\bar{e}$ out of 8.9 million can tunnel through the 10.0 eV barrier on the average.

For 2.0 eV, $T_2 = 2.4 \times 10^{-7}$. For 2.0 eV

So, twice as likely correspond to a 1.0 eV.

(b) If the barrier is doubled in width to 1.0 nm,

$$T_1'(1.0\,eV) = 1.3 \times 10^{-14},$$

$$T_2'(2.0\,eV) = 5.1 \times 10^{-14}.$$

Evidently T is more sensitive to the width of the barrier than the particle energy.